Temperature sensor

DETECT | PICO | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling



Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer



Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import TemperatureSensor

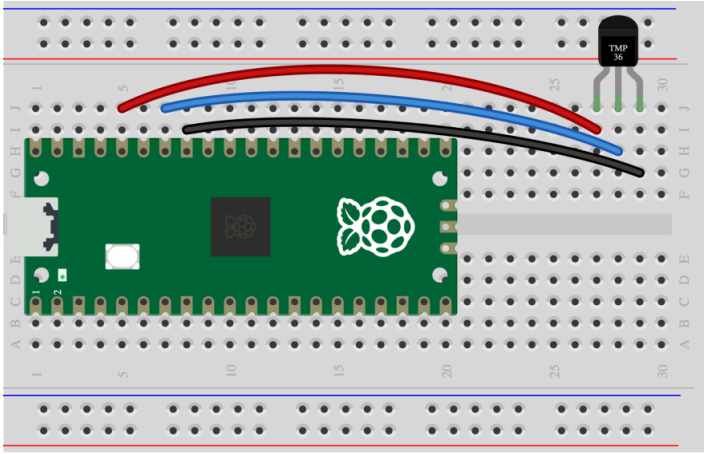
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(28, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

Wiring

PICO



TMP36 **+Vs** → 3V3 OUT (pin 36), **Vout** → GP28 (pin 34), **GND** → GND (pin 33)